

Philip Andrew White

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Education

- Doctor of Philosophy: Statistical Science; Duke University; 2019
 - Dissertation: Topics in Bayesian Spatiotemporal Prediction of Environmental Exposure
 - Committee: Alan Gelfand (chair), Fan Li (co-chair), Colin Rundel, and Ben Goldstein
- Master of Science: Statistics; Brigham Young University; 2015
 - Thesis: Bayesian Gaussian Process Model for Antarctic Surface Mass Balance and Proposing New Field Measurements
 - Committee: C. Shane Reese (chair), William F. Christensen, and Shannon Tass
- Bachelor of Science: Applied Physics; Brigham Young University; 2014
 - Graduated *Magna Cum Laude*; speaker at college graduation
 - Senior Thesis: Bayesian Model for Antarctic Surface Mass Balance
 - Minors: Mathematics, Scandinavian Studies

Professional Experience

Brigham Young University

- Assistant Professor *May 2025 – Present*
- Assistant Professor (on leave) *May 2023 – May 2025*
- Assistant Professor *April 2019 – May 2023*

Berry Consultants

- Statistical Scientist *May 2023 – May 2025*

The Climate Corporation

- Data Science Intern: Disease Risk Modeling *May 2018 – Aug 2018*

Statistical Consulting Experience

- Philip White Statistical Consulting LLC, Provo, UT; 2025 – Present
- Department of Clinical Sciences, North Carolina State University, College of Veterinary Medicine; Raleigh, NC; 2019 – Present
- Hare Lab, Duke University, Department of Evolutionary Anthropology; Durham, NC; 2017 – Present
- Berry Consultants; 2023 – 2025
- Arbinger Institute; Farmington, Utah; 2019 – 2020

Peer-Reviewed Publications

- P.G. Bissiri, B. Nipoti, E. Porcu, and **P. White** (2025+), “A Nonparametric Framework for Correlation Functions on Product Metric Spaces,” accepted at *International Statistical Review*.
- R.M.P. Caddiell, **P.A. White**, E.H. McNamee, A.M. Lynch, B.D.X. Lascelles, M.E. Gruen (2025), “Tailoring Treatment: Dog Breed Status Influences Pain Assessment and Treatment in Emergency Veterinary Care,” *Frontiers in Pain Research*, 6, 1589082.
- M. Ferrans, H. Salomons, K. Moore, **P. White**, B. Hare, and M. E. Gruen (2025), “Salivary cortisol is an unreliable correlate of serum cortisol in adult pet dogs and assistance dog puppies,” *Scientific Reports*, 15, 15986. [link](#)
- G. Cleanthous, A. G. Georgiadis, and **P. White** (2025), “Pointwise density estimation on metric spaces and applications in seismology,” *Metrika*, 88, 119–148. [link](#)

- **P. White**, A.E. Gelfand, H. Frye, J. Slingsby, and J. Silander Jr (2024), “Good Modelling Practice in Ecology, the Hierarchical Bayesian Perspective,” *Ecological Modelling*, 496, 110847. [link](#)
- *A. Alegría, G. Cleanthous, A. Georgiadis, E. Porcu, and **P. White** (2024), “Gaussian random fields on the product of spheres: theory and applications,” *Electronic Journal of Statistics*, 1, 1394-1435. [link](#)
- **P. White**, H. Frye, J. Slingsby, J. Silander Jr, and A.E. Gelfand, (2024), “Generative Spatial Generalized Dissimilarity Mixed Modeling (spGDMM): an enhanced approach to modelling beta diversity,” *Methods in Ecology and Evolution (MEE)*, 15(1), No. 1, 214-226. [link](#)
- M. Heaton, B.K. Dahl, C. Dayley, R.L. Warr, and **P. White** (2024), “Integrating Machine Learning and Bayesian Nonparametrics for Flexible Modeling of Point Pattern Data,” *Computational Statistics & Data Analysis*, 191, 107875. [link](#)
- E. Porcu, **P. White**, and M. Genton (2023), “Stationary Non-Separable Space-Time Covariance Functions on Networks,” *Journal of the Royal Statistical Society: Series B*, 85(5), 1417–1440. [link](#)
- **P. White**, M. Christensen, H. Frye, A. Gelfand, and J. Silander Jr (2023), “Joint Multivariate and Functional Modeling for Plant Traits and Reflectances,” *Environmental and Ecological Statistics*, 30, 501-528. [link](#)
- R.M.P. Caddiell, **P. White**, B.D.X. Lascelles, K. Royal, K. Ange van-Heugten, M.E. Gruen (2023). “Veterinary education and experience shape beliefs about dog breeds Part 1: Pain sensitivity,” *Nature Scientific Reports*, 13, 13846. [link](#)
- R.M.P. Caddiell, **P. White**, B.D.X. Lascelles, K. Royal, K. Ange van-Heugten, M.E. Gruen (2023). “Veterinary education and experience shape beliefs about dog breeds Part 2: Trust,” *Nature Scientific Reports*, 13, 13847. [link](#)
- R. Caddiell, R.M. Cunningham, **P.A. White**, B. Duncan, X. Lascelles and M.E. Gruen (2023), “Pain Sensitivity Differs Between Dog Breeds but Not in the Way Veterinarians Believe,” *Frontiers in Pain Research*, 4, 1165340. [link](#)
- H. Salomons, K.C.M. Smith, M. Callahan-Beckel, M. Callahan, K. Levy, B.S. Kennedy, B.E. Bray, G.E. Gnanadesikan, D.J. Horschler, M. Gruen, J. Tan, **P. White**, B.M. vonHoldt, E.L. MacLean, and B. Hare (2023), “Response to Hansen Wheat et al.: Additional analysis further supports the early emergence of cooperative communication in dogs compared to wolves raised with more human exposure,” *Learning & Behavior*, 51, 131-134. [link](#)
- A.J. Mason, J.M. McCardell, **P.A. White**, and J.S. Colton (2023), “Improving performance in upper-division electricity and magnetism with explicit incentives to correct mistakes,” *Physical Review - Physics Education Research*, 19(2), 020104. [link](#)
- D. Sheanshang[†], **P. White**, and D. Keeler (2023), “Outlier Accommodation with Semiparametric Density Processes: A Study of Antarctic Snow Density Modelling,” *Statistical Modelling*, 23(2), 151-172. [link](#)
- L. Warr[†], M. Heaton, W. Christensen, **P. White**, and S. Rupper (2023), “Distributional Validation of Precipitation Data Products with Spatially Varying Mixture Models,” *Journal of Agricultural, Biological and Environmental Statistics*, 28(1), 99-116. [link](#)
- M. Heiner, M. Heaton, B. Abbott, **P. White**, C. Minaudo, and R. Dupas (2023), “Model-based clustering of trends and cycles of nitrate concentrations in Rivers across France,” *Journal of Agricultural, Biological and Environmental Statistics*, 28(1), 74-98. [link](#)
- **P. White**, D. Keeler, D. Sheanshang[†], and S. Rupper (2022), “Improving Piecewise Linear Snow Density Models through Hierarchical Spatial and Orthogonal Functional Smoothing,” *Environmetrics*, 33(5), e2726. [link](#)
- X. Emery, E. Porcu, and **P. White** (2022), “New Validity Conditions for the Multivariate Matérn Coregionalization Model, with an Application to Exploration Geochemistry,” *Mathematical Geosciences*, 54, 1043–1068. [link](#)

- **P. White**, H. Frye, M. Christensen, A. Gelfand, and J. Silander Jr (2022), “Spatial Functional Data Modeling of Plant Reflectances,” *Annals of Applied Statistics*, 16(3), 1919-1936. [link](#)
- E. Porcu and **P. White** (2022), “Random Fields on the Hypertorus: Covariance Modeling and Applications,” *Environmetrics*, 33(1), e2701. [link](#)
- A. Bowie, W. Zhou, J. Tan, **P. White**, T. Stoinski, S. Yangjie, and B. Hare (2022), “Motivating children’s cooperation to conserve forests,” *Conservation Biology*, 36(4), e13922. [link](#)
- G. Cleanthous, E. Porcu, and **P. White** (2021), “Regularity and Approximation of Gaussian Random Fields Evolving Temporally over Two-Point Homogeneous Spaces,” *TEST*, 30(4), 836-860. [link](#)
- H. Salomons, K. Smith, M. Callahan-Beckel, M. Callahan, K. Levy, B. Kennedy, E. Bray, G. Gnanadesikan, D. Horschler, M. Gruen, J. Tan, **P. White**, E. MacLean, and B. Hare (2021), “Cooperative Communication with Humans Evolved to Emerge Early in Domestic Dogs,” *Current Biology*, 31(14), 3137-3144. [link](#)
- **P. White**, D. Keeler, and S. Rupper (2021), “Hierarchical Integrated Spatial Process Modeling of Monotone West Antarctic Snow Density Curves,” *Annals of Applied Statistics*, 15(2), 556-571. [link](#)
- *A. Alegría, P.G. Bissiri, G. Cleanthous, E. Porcu, and **P. White** (2021), “Multivariate Isotropic Random Fields on Spheres: Nonparametric Bayesian Modeling and L^p -Fast Approximations,” *Electronic Journal of Statistics*, 15(1), 2360-2392. [link](#).
- **P. White** and A. Gelfand (2021), “Generalized Evolutionary Point Processes: Model Specifications and Model Comparison,” *Methodology and Computing in Applied Probability*, 23, 1001–1021. [link](#).
- **P. White** and A. Gelfand (2021), “Multivariate Functional Data Modeling with Time-varying Clustering,” *TEST*, 30(3), 586–602. [link](#)
- M. Gruen, **P. White**, and B. Hare (2020), “Do Dog Breeds Differ in Pain Sensitivity? Veterinarians and the Public Believe They Do,” *PLoS ONE*, 15(3), e0230315. [link](#).
- **P. White**, C.S. Reese, W. Christensen, and S. Rupper (2019), “A Model for Antarctic Surface Mass Balance and Ice Core Site Selection,” *Environmetrics*, 30(8), e2579. [link](#).
- **P. White** and E. Porcu (2019), “Towards a Complete Picture of Stationary Covariance Functions on Spheres Cross Time,” *Electronic Journal of Statistics*, 13(2), 2566-2594. [link](#).
- **P. White**, A. Gelfand, E. Rodrigues, and G. Tzintzun (2019), “Pollution State Modeling for Mexico City,” *Journal of the Royal Statistical Society - Series A*, 182(3), 1039-1060. [link](#).
- **P. White** and E. Porcu (2019), “Nonseparable Covariance Models on Circles Cross Time: A Study of Mexico City Ozone,” *Environmetrics*, 30(5), e2558. [link](#)
- **P. White**, C. Berrett, S. Tass, and M. Findlay (2019), “Modeling Efficiency of Foreign Aid Allocation in Malawi,” *The American Statistician*, 73(4), 385-399. [link](#)
- **P. White**, A. Gelfand, and T. Utlaut (2017), “Prediction and model comparison for areal unit data,” *Spatial Statistics*, 22(1), 89-106. [link](#)
- J. S. Colton, D. Meyer, K. Clark, D. Craft, J. Cutler, T. Park, and **P. White** (2012), “Long-Lived electron spins in a modulation doped (100) GaAs quantum well,” *Journal of Applied Physics*, 112(8), 084307. [link](#)

* Under the request of one or more coauthors, the authors are listed in alphabetical order.

† Mentored Student co-author

Under Review

- P.A. White, N.T. James, D.A. Berry, V. Manax, L. Tracey, and A.C. Stubbs, “Bayesian Hierarchical Modeling for Personalized Dose Escalation and De-escalation”
- P. White, L. Berry, M. Quintana, N. Johnson, et al. “Platform Trials for Efficient Treatment Discovery in Myotonic Dystrophy Type 1.”
- An early-phase ALS trial using natural history data as a control. Title and authors redacted for confidentiality.

In Preparation

- P.Wang, M. Quintana, P. White, and M. Vestrucci, “Leveraging External Controls from PRO-ACT for Exploratory Efficacy Assessment in Early-Phase ALS Trials.”
- P. White, H. Frye, J. Slingsby, J. Silander Jr, and A.E. Gelfand, “Dynamic Spatial Generalized Dissimilarity Mixed Modeling: a dynamic spatial beta diversity model.
- H. Horsley, M.O. Alver, I. Ellingsen, and P. White, “Modelling marine beta diversity with a hierarchical Bayesian spatial model: Application of spGDMM to simulated plankton communities.”

Other Publications

- P. White (2019), “Topics in Bayesian Spatiotemporal Prediction of Environmental Exposure,” Ph.D. Dissertation, Duke University.
- P. White (2015), “Bayesian Gaussian Process Model for Antarctic Accumulation and Proposing New Field Measurement,” Masters Project, Brigham Young University.
- P. White (2014), “ORCA final report,” *2014 Journal of Undergraduate Research* [Online], Brigham Young University.
- P. White (2014), “Quantifying Climate Change: Bayesian Model for Antarctic Surface Mass Balance,” Senior Thesis, Brigham Young University.

Posters and Presentations

- Brigham Young University Department of Statistics Seminar (2025), “Bayesian Hierarchical Modeling for Personalized Dose Escalation and De-escalation.”
- ASA Biopharmaceutical Section Regulatory-Industry Statistics Workshop (2025), Invited Talk, “Adaptive Sample Size Re-estimation with Longitudinal Endpoints.”
- ENCALS meeting (2024), Contributed Poster, “Using Historical Control Data in Early-Stage Exploratory Efficacy Analysis of Monepantel in ALS.”
- The 25th International Conference on Computational Statistics (2023), Invited Talk, “Multivariate isotropic random fields on spheres: Nonparametric Bayesian modeling and L_p fast approximations.”
- 13th International Conference of the ERCIM WG on Computational and Methodological Statistics (CMStatistics 2022), Invited Talk, “Scalable Warped Directional Traffic Network Models for Traffic Accident Data.”
- Berry Consultants, Research Seminar Series (2022), “Hierarchical Integrated Spatial Process Modeling of Monotone West Antarctic Snow Density Curves.”
- ASA ENVR Workshop (2022), Contributed Poster, “Spatial Functional Data Modeling of Plant Reflectances.”
- Joint Statistical Meetings (2022), Topic-Contributed Talk, “Spatial Functional Data Modeling of Plant Reflectances.”
- Quality & Productivity Research Conference (2022), Invited Talk, “Some Challenges in Spatial Functional Data Analysis of West Antarctic Snow Density.”
- National University of Ireland, Maynooth (2021), Invited Talk, “Some Challenges in Spatial Functional Data Analysis of West Antarctic Snow Density.”
- ICSA Applied Statistics Symposium (2021), Invited Talk, “Hierarchical Integrated Spatial Process Modeling of Monotone West Antarctic Snow Density Curves.”
- Joint Statistical Meetings (2021), Topic-Contributed Talk, “Hierarchical Integrated Spatial Process Modeling of Monotone West Antarctic Snow Density Curves.”
- The International Environmetrics Society (TIES) Virtual Conference (2020), Invited Talk, “A Model for Antarctic Surface Mass Balance and Ice Core Site Selection.”
- Brigham Young University Department of Statistics Seminar (2020), “Hierarchical Integrated Spatial Process Modeling of Monotone West Antarctic Snow Density Curves.”

- Joint Statistical Meetings (2020), Contributed Talk, “Hierarchical Integrated Spatial Process Modeling of Monotone West Antarctic Snow Density Curves.”
- New England Statistics Symposium (2019), Contributed Talk, “Multivariate Functional Data Modeling with Time-varying Clustering.”
- Joint Statistical Meetings (2019), Contributed Talk, “Multivariate Functional Data Modeling with Time-varying Clustering.”
- Brigham Young University Department of Statistics Seminar (2018), Invited Talk, “Nonseparable Covariance Models on Circles Cross Time: A Study of Mexico City Ozone.”
- The RAND Corporation (2018), Invited Talk, “Nonseparable Covariance Models on Circles Cross Time: A Study of Mexico City Ozone.”
- Los Alamos National Labs Seminar (2018), Invited Talk, “Nonseparable Covariance Models on Circles Cross Time: A Study of Mexico City Ozone.”
- Facebook Research Labs (2018), Invited Talk, “Nonseparable Covariance Models on Circles Cross Time: A Study of Mexico City Ozone.”
- International Conference on Advances in Interdisciplinary Statistics and Combinatorics (AISC) (2018), Contributed Talk, “Nonseparable Covariance Models on Circles Cross Time: A Study of Mexico City Ozone.”
- ASA ENVR Workshop - Statistics for the Environment: Research, Practice and Policy (2018), Contributed Poster, “Pollution State Modeling for Mexico City.”
- Society of Duke Fellows (2017), Invited Talk, “Prediction and Model Comparison for Areal Unit Data.”
- American Geophysical Union (2017), New Orleans, LA., Contributed Poster, “A Model for Antarctic Surface Mass Balance and Ice Core Site Selection.”
- Conference on Data Analysis (2014), Poster, Santa Fe, NM., Contributed Poster, “A Model for Antarctic Surface Mass Balance and Ice Core Site Selection.”

Courses Taught

- Instructor of Record
 - Brigham Young University
 - Statistics 250 (R Programming): Fall 2025
 - Statistics 330 (Statistical Modeling 2): Winter 2020, Winter 2021, Winter 2023, Fall 2025
 - Statistics 437 (Biostatistics): Winter 2026
 - Statistics 641 (Probability Theory & Mathematical Statistics 1): Fall 2019, Fall 2020, Fall 2021, Fall 2022
 - Swedish 201: Winter 2012
 - Swedish 202: Fall 2011, Fall 2012
 - Duke University
 - Statistics 111 (Probability and Statistics): Summer I 2017
- Teaching Assistant
 - Duke University
 - Statistics 322/522 (Design of Surveys and Causal Studies): Spring 2019
 - Statistics 944 (Spatial Statistics): Fall 2018
 - Statistics 444/644 (Spatio-temporal Modeling): Spring 2018, Fall 2018
 - Statistics 532 (Theory of Inference): Fall 2017
 - Statistics 111 (Probability and Statistics): Spring 2017

Awards

- 2021-2025, “Point Process Models for Traffic Risk Analysis and Crash Prevention, the National Science Foundation, 2020 (Role: Co-PI, Amount: \$199,941).
- 2021-2022, “The Role of Temperature Variation for Reconstructing the Advance and Retreat of Glacial Ice using Thermal and Radar Imaging,” Brigham Young University Interdisciplinary Research (IDR) Origination Awards, 2020 (Role: Co-PI, Amount: \$119,910).
- The 2020 BYU Department of Statistics Young Scholar Faculty Fellowship (Amount: \$6,000)
- 2019 Wiley-TIES Best Environmetrics Paper Award for “A model for Antarctic surface mass balance and ice core site selection,” (Amount: Travel award and \$750).
- 2019 STATMOS Workshop - Young Researcher Travel Award, 2019 (Amount: \$1,500)
- 2015-2019, James B. Duke Fellowship Recipient, Duke University, (Amount: \$20,000)
- 2018 ENVR Workshop - Statistics for the Environment: Research, Practice and Policy, Student travel award, (Amount: \$1,500)
- 2018, Department of Statistical Science TA of the Year (Amount: \$1,500)
- 2015-2016, Statistical Science First Year Fellowship, Duke University
- 2014, Conference on Data Analysis Student Travel Award, (Amount: \$1,000)
- 2014-2015, Brigham Young University Student Research Conference Session Winner

Service

- Student Mentoring

Year Graduated	Student	Role
—	Jonas Schiel	Research Mentor
—	Caleb Christensen	MS Committee Chair
—	Ty Hawkes	MS Committee Member
—	Jackson Passey	MS Committee Member
2025	Jason Meziere	Ph.D. Committee (Physics)
2024	Daniel Smith	Research Co-Mentor
2024	Benjamin Dahl	Research Co-Mentor
2023	Caleb Dayley	MS Committee Chair
2023	Sam Spackman	Research Mentor
2023	Joshua Christensen	MS Committee Member
2023	Jace Ritchie	MS Committee Member
2023	Colin Free	MS Committee Member
2022	Carly Lundgreen	Research Co-Mentor
2021	Daniel Sheanshang	MS Committee Chair
2021	Lynsie Warr	MS Committee Member
2021	Madeline Morris	MS Committee Member
2020	Maryanne Allen	Research Mentor
2020	Derik Mehl	Research Mentor
2020	Spencer Ebert	MS Committee Member
2020	Shelby Taylor	MS Committee Member

- Departmental Service
 - Graduate Admissions (2025 -)
 - Alumni/Employer Relations (2025 -)
 - Curriculum (2025 -)
 - Student Outreach (2022 - 2023)

- Rank and Status Revision Committee (2022 - 2023)
- Recruiting and Retention Committee (2021 - 2022)
- Comprehensive Exam Committee (2020 - 2023)
- Seminar Co-Coordinator (2019 - 2021)
- To Profession
 - Associate Editor: *Environmetrics*.
 - Scientific Planning Committee for CMStatistics 2026.
 - Organized or co-organized sessions for statistical conferences – JSM 2020, CMStatistics 2021, QPRC 2022, CMStatistics 2022, CMStatistics 2026.
 - Keynote speaker for Brigham Young University Mu Sigma Rho induction, 2021.
 - Refereed papers for the following journals: *Annals of Applied Statistics*; *Biometrics*; *Computational Statistics and Data Analysis*; *Environmental and Ecological Statistics*; *Electronic Journal of Statistics*; *Environmetrics*; *Journal of Agricultural, Biological and Environmental Statistics*; *Journal of the American Statistical Association*; *Journal of the Quantitative Analysis of Sports*; *Mathematical Inequalities & Applications*; *Spatial and Spatio-temporal Epidemiology*; *Spatial Statistics*; *Statistics and Computing*; *Stochastic Environmental Research and Risk Assessment*.

Memberships

- American Statistical Association: 2014-Present
- International Society for Bayesian Analysis: 2017-Present
- The International Environmetrics Society (TIES): 2021-Present